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PERFORMANCE SPECIFICATION

DISTRIBUTION BOX, VEHICLE

This specification is approved for use by U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a Vehicle Distribution Box, referred to as the VDB.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 6110

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2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

12328513 - Distribution Box, Vehicle (1A1) (Interface).

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

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(Application for copies should be addressed to American National Standard Institute (ANSI), 11 West 42nd Street, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4.

3.2 Materials. Materials shall be in accordance with the manufacturer's materials specifications for VDBs. Materials shall be capable of meeting all the operational and environmental requirements specified herein. Asbestos and cadmium materials shall not be used in any form in any part of the VDB. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcuries per gram or activity per item equals or exceeds 0.01 microcuries (see 4.7.1).

3.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals which would encourage galvanic action shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces (see 4.7.1).

3.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 4.7.1).

3.3 Design and construction. Components and subassemblies shall be designed and constructed in accordance with the VDB manufacturer's standards. VDBs shall conform to the envelope dimensions, and the electrical and mounting interfaces as shown in Drawing 12328513 (see 4.7.1 and 4.7.2).

3.3.1 Cementing of gasket. Cementing of gasket shall be in accordance with the manufacturer's commercial practices (see 4.7.1).

3.3.2 Lockwire. Lockwire shall be in accordance with the commercial practices (see 4.7.2).

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3.3.3 Environmental Stress Screening (ESS). Unless otherwise specified (see 6.2), the VDB shall be subjected to ESS (see 6.4.2), in accordance with Drawing 12328513 (see 4.7.1).

3.4 Performance characteristics.

3.4.1 Primary power source. Power for the VDB shall be supplied from a 28 volt (V) direct current (dc) source having the following characteristics and connections (see 4.7.3):

a. Characteristics:
   1. Upper limit - 30. 0 Vdc.
   2. Nominal - 28.0 Vdc.
   3. Lower limit - 18.0 Vdc.

b. Connections:
   1. 28 Vdc connection (as specified).
   2. Power ground connection - E1.

NOTE: Unless otherwise specified, 28 Vdc refers to nominal 28 Vdc. Unless otherwise specified, the negative side of all power sources shall be connected to E1.

3.4.2 Relay switching functions. Vehicle functions shall be controlled by the following relays:

a. Cold start relay.
b. Fire suppressant relay.
c. Starter cutout relay.
d. Lamp test relay.
e. Slave power relay.
f. Hydraulic pump motor relay.

3.4.2.1 Cold start relay, K1. When 28 Vdc is applied to J10-A, and J8-E is connected to E1, the voltage at J7-A and J7-B shall be within 0.5 Vdc of the source voltage (see 4.7.4.1).

3.4.2.2 Fire suppressant relay, K8. Operation of K8 relay shall be as follows (see 4.7.4.2):

a. When 28 Vdc is applied to J2-J, J8-BR is connected to ground through a 1.0 kohm, 5 percent (%), 1/4 Watt (W) resistor and toggle switch S2 is momentarily activated, the voltage at J8-BR shall be less than 0.5 Vdc.
b. When 28 Vdc is applied to J2-J and a 10.0 $\pm$ 1.0 Vdc, 125 $\pm$ 50 millisecond (ms) pulse is applied between J1-C (plus (+)) and E1 (minus (-)), the relay shall latch as indicated by the constant 28 Vdc source voltage at J8-BR.

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3.4.2.3 Starter lockout relay, K3. When 28.0 to 30.0 Vdc is applied to J8-AA, the following shall occur (see 4.7.4.3):

a. There shall be continuity between J8-BC and E1 and between J15-L and E1 when a ground is applied to J7-M.
b. There shall be continuity between J8-BC and E1 and between J15-L and E1 when the frequency of the following sine wave signal, applied between J7-M and J7-N, is less than 575 hertz (Hz). Continuity shall not exist between J8-BC and E1 and between J15-L and E1 when the frequency of the following sine wave signal, applied between J7-M and J7-N, is greater than 1100 Hz:

    1. Signal amplitude: 2.5 $\pm$ 0.5 V peak-to-peak (pp).
    2. Source impedance: greater than 100 ohms.

3.4.2.4 Lamp test relay, K4. When 28 Vdc is applied to J8-AF, J8-AA, and J10-A, the following shall occur (see 4.7.4.4):

a. Headphone warning signal shall occur (see 3.4.4.1).
b. Flasher indicator warning signal shall occur (see 3.4.4.2).

3.4.2.5 Master/slave power transfer relay, K5. Operation of K5 shall be as follows (see 4.7.4.5):

3.4.2.5.1 Slave, proper polarity. When 28 Vdc is applied between J13 (inner socket) (+) and J13 (outer shell) (-), the following shall occur:

a. J10-A shall be within 0.5 Vdc of the 28 Vdc source.
b. The green portion of "Slave Receptacle Power" light-emitting diode (LED) shall illuminate.

3.4.2.5.2 Slave, reverse polarity. When ground is removed from E1 and 28 Vdc is applied between J13 (inner socket) (-) and J13 (outer shell) (+), the following shall occur:

a. Continuity shall not exist between J10-A and J13 (inner socket).
b. The red portion of the "Slave Receptacle Power" LED shall illuminate.

3.4.2.5.3 Vehicle power bus, normal. When 28 Vdc is applied to J10-A and the "Energize Slave Receptacle" pushbutton switch is depressed, the inner socket of J13 shall be within 0.5 Vdc of the 28 Vdc source.

3.4.2.6 Hydraulic pump motor relay, K7. When 28 Vdc is applied between J8-U (+) or J3-C (+) and J8-BD (-), there shall be continuity between J10-A and J17-A and between J10-A and J15-V (see 4.7.4.6).

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3.4.3 Nuclear-Biological-Chemical (NBC) system operation. When 28 Vdc is applied to J10-A and the "NBC system" toggle switch is in the "ON" position, the associated LED shall be illuminated and 28 Vdc shall be present at J1-D, J1-E, J1-F and J1-G (see 4.7.5).

3.4.4 Warning signal functions. When 28 Vdc is applied to J8-AA and any of the following occurs, the headphone warning signal and the flashing indicator signal shall occur (see 4.7.6):

a. Ground is applied to any one of the following, J4-X, J7-R, J7-S, J7-T, J7-U, J7-a, and J7-c.
b. 28 Vdc is applied to J8-X through a resistance of 1000 to 4700 ohms and ground is applied to J7-h.
c. The lamp test relay K4 is energized by applying 28 Vdc to J8-AF and J10-A.

3.4.4.1 Headphone warning signal. The headphone warning signal shall be a 760 ± 150 Hz voltage signal interrupted at the rate of 1.5 ± 0.5 Hz by the flashing indicator signal in 3.4.4.2. The signal shall be present at J8-C from the time a warning signal function is applied until a ground is momentarily applied to J8-AC. When a 185 ± 80 kohm, 1/4 W resistor is connected between J8-C and E1, the shape of the 760 Hz signal shall be similar to that shown in figure 1.

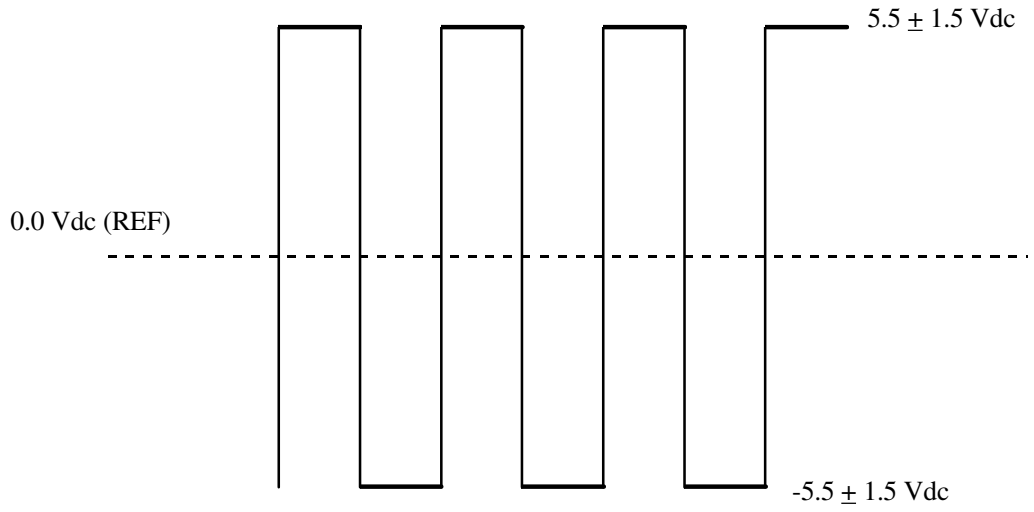
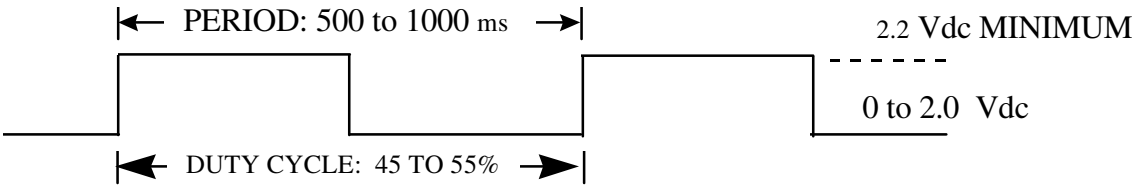


FIGURE 1. Headphone warning signal.

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3.4.4.2 Flashing indicator signal. The flashing indicator signal shall be a voltage pulse train as shown in figure 2. The signal shall be present at J8-AJ only during the presence of a warning signal function.



NOTE: Value shown is for nominal 28.0 Vdc applied to J8-AA. For source Vdc (MIN) (18.0 Vdc), peak voltage shall be +12.0 Vdc minimum. For source Vdc (MAX) (30.0 Vdc), peak voltage shall be no greater than 28.0 Vdc.

FIGURE 2. Flashing indicator signal.

3.4.5 Vehicle motion switching function. When 18.0 to 26.0 Vdc is applied between J4-W and E1, and a 1800 ohm ± 5%, 1/2 W resistor is connected between J4-K and E1, and a 40.0 ± 2.0 Hz sine wave input signal is applied between J9-N and J9-M, and voltage at J4-K shall be as follows (see 4.7.7):

a. No greater than 2.0 Vdc when the input signal has an amplitude of less than 2.6 V root mean square (rms).
b. At least 10.5 Vdc when the input signal has an amplitude of 9.2 ± 0.1 V rms.

3.4.6 STE M1/FVS and Test Measurement Diagnostic Equipment (TMDE) functions. The STE M1/FVS and TMDE functions shall be as follows:

a. Resistance measurements.
b. Direct current voltage.
c. Top dead center signal.
d. Air cleaner circuit.

3.4.6.1 Resistance measurements. Resistance between connector pins shall be specified in table I (see 4.7.8.1).

TABLE I. Resistance measurements.

From connector pin	To connector pin	Resistance (ohms)
J14-h	J14-j	124 to 130
J15-h	J15-j	1085 to 1115

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3.4.6.2 Direct-current voltage. When a voltage source of not more than 40 Vdc is applied between J16-a(+) and J16-b(-), and between J16-W(+) and J16-X(-), the voltage between the pins of the connectors listed in table II shall be a fraction of the voltage source being applied, as shown in table II (see 4.7.8.2).

TABLE II. DC voltage.

From connector pin	To connector pin	Fraction of voltage source
J16-X	J15-D	0.1144 to 0.1272
J16-X	J15-Y	0.475 to 0.525
J16-X	J15-p	0.475 to 0.525
J16-X	J15-x	0.475 to 0.525
J16-X	J15-z	0.475 to 0.525
J16-X	J15-AB	0.475 to 0.525
J16-X	J15-AD	0.475 to 0.525
J16-b	J14-D	0.1144 to 0.1272
J16-b	J14-AD	0.475 to 0.525

3.4.6.3 Top dead center (TDC) signal. When the following inputs are present, the resultant signal at J15-c shall be as a 6 to 20 ms pulse, as shown in figure 3, with an amplitude within 1.5 Vdc of the voltage source used in b. below (one pulse shall be generated for each negative-going transition of the input signal at J16-G) (see 4.7.8.3).

a. 28 Vdc is applied to J8-AA.
b. A voltage source between 5.0 and 12.0 Vdc is applied to J14-c through a resistor that will limit the current to 2.2 ± 0.3 milliamperes (mA).
c. A 2.0 ± 0.5 Vdc peak-to-peak (p-p) square-wave signal symmetrical with respect to ground, with a frequency from 1 to 20 Hz applied to J16-G.

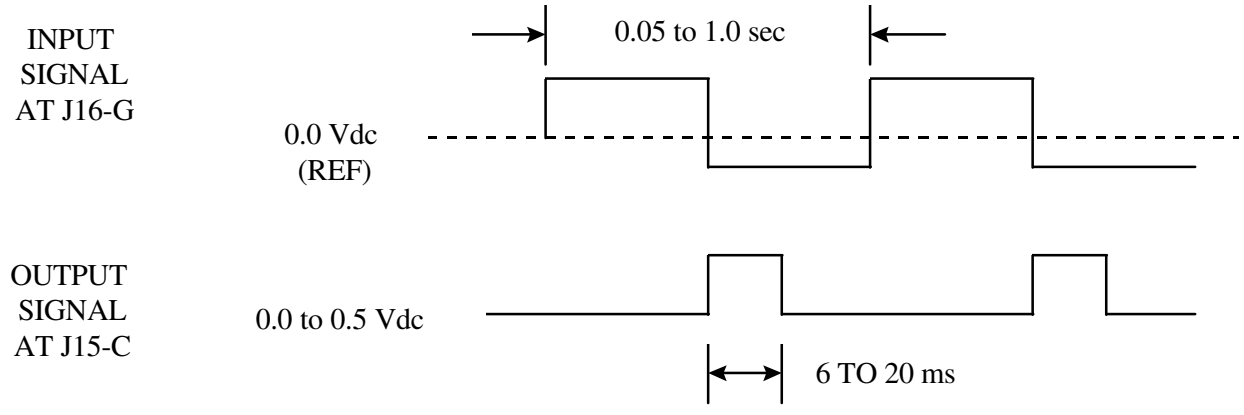


FIGURE 3. Top dead center signal.

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3.4.7 Coolant low-level sending circuit. When 28 Vdc is applied to J8-AA, the voltage at J7-k shall be within 1.0 Vdc of the source voltage (see 4.7.9).

3.4.8 Master power relay current-limiting function. When 28 Vdc is applied to J10-A or J2-H, the voltage at J8-Z shall be equal to the source voltage minus 3.5 ± 1.5 Vdc at a load current of 450 ± 50 mA. The short-circuit load current from J8-Z shall be limited to 0.70 to 1.35 amperes (A) (see 4.7.10).

3.4.9 Indicator lamp control functions. The following are the indicator lamp functions:

a. Lamp test
b. Front bilge pump
c. Rear bilge pump
d. Smoke screen
e. Cold start
f. Ramp unlock
g. Fuel filter restriction
h. Engine oil low pressure
i. Transmission oil temperature
j. Transmission oil pressure
k. Launcher up
l. Air cleaner restriction
m. Coolant high temperature
n. Fire suppression
o. Turret power on

3.4.9.1 Lamp test. When 28 Vdc is applied to J8-AA and J8-AF, the voltage at the following connector pins shall be from 1.4 to 2.8 Vdc (see 4.7.11.1):

a. J7-U
b. J7-S
c. J7-R
d. J4-X
e. J7-a
f. J7-c
g. J7-T

3.4.9.2 Diode isolation. Diode isolation shall exist between connector pins as specified in table III (see 4.7.11.2).

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TABLE III. Diode isolation.

Anode	Cathode
J8-AY	J8-AE
J9-D	J8-AE
J8-BE	J8-H
J3-A	J8-H
J8-B	J8-AU
J7-P	J8-AU
J8-F	J8-E
J8-AK	J7-c
J8-AS	J7-R
J8-AP	J7-T
J8-AN	J7-U
J8-AR	J4-X
J8-W	J7-a
J8-AB	J7-S

3.4.9.3 Lamp test circuits. When 28 Vdc is applied to J8-AF, each lamp test circuit specified in table IV shall be capable of maintaining a current flow of at least 10 mA when a current limited (110 mA maximum), forward biasing dc voltage (not exceeding the 28 Vdc bus) is applied across the connector pins as specified (see 4.7.11.3).

TABLE IV. Lamp test diode isolation.

Anode	Cathode
J10-A	J8-AE
J10-A	J8-H
J10-A	J8-AU
J8-F	E1
J8-T	E1
J3-X	E1
J8-a	J8-BP
J8-a	J8-BR

3.4.10 Ramp control function. Diode isolation shall exist between J8-BD and J8-BT (anode connected to J8-BD) (see 4.7.12).

3.4.11 Wiring continuity. Wiring continuity shall be in accordance with table V (see 4.7.13).

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TABLE V. Continuity.

From connector pin	To connector pin	From connector pin	To connector pin
E1	J1-H	J3-H	J8-BT
E1	J3-E	J3-H	J15-S
E1	J6-A	J3-K	J4-B
E1	J6-D	J3-K	J11-E
E1	J6-E	J3-K	J11-F
E1	J6-K	J3-K	J11-K
E1	J6-L	J3-K	J11-L
E1	J6-S	J3-L	J8-AD
E1	J6-V	J4-D	J3-P
E1	J7-F	J4-E	J3-R
E1	J7-G	J4-F	J11-C
E1	J8-D	J4-j	J6-M
E1	J11-B	J4-L	J11-G
E1	J14-AB	J4-W	J3-Q
E1	J14-F	J4-W	J8-BB
E1	J15-F	J4-W	J8-AZ
E1	J16-d	J4-W	J11-D
E1	J16-H	J4-Z	J6-R
J1-B	J8-BS	J5-A	J4-V
J1-D	J1-E	J5-A	J12-A
J2-B	J14-X	J5-B	J4-S
J2-C	J14-W	J5-B	J12-B
J2-C	J14-Y	J5-C	J4-J
J2-C	J15-W	J5-C	J12-C
J2-J	J2-K	J5-D	J4-M
J5-D	J12-D	J8-AZ	J8-a
J5-E	J4-H	J8-BA	J3-d
J5-E	J12-E	J8-BC	J15-L
J5-F	J4-U	J8-BE	J3-A
J5-F	J12-F	J8-BF	J3-B
J5-G	J4-N	J8-BH	J3-b
J5-G	J12-G	J8-BJ	J3-a
J5-H	J4-P	J8-BK	J3-c
J5-H	J12-H	J8-BL	J9-E
J5-J	J4-R	J8-BM	J3-Z
J5-J	J12-J	J8-BN	J11-A
J5-K	J4-T	J8-BN	J14-E
J5-K	J12-K	J8-BN	J14-V
J5-L	J4-b	J8-BN	J15-E
J5-L	J12-L	J8-BU	J9-F
J5-M	J4-c	J8-BV	J9-J
J5-M	J12-M	J8-J	J9-B
J5-N	J4-G	J8-L	J9-G

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TABLE V. Continuity - Continued.

From connector pin	To connector pin	From connector pin	To connector pin
J5-N	J12-N	J8-T	J3-X
J5-P	J4-a	J8-U	J3-C
J5-P	J12-P	J8-Y	J2-L
J6-C	J6-P	J9-H	J8-AL
J6-F	J6-N	J10-A	J3-L
J6-G	J8-M	J10-A	J3-N
J6-H	J6-J	J10-A	J6-X
J6-H	J6-W	J10-A	J8-a
J6-X	J6-B	J10-A	J8-N
J6-X	J6-U	J10-A	J8-P
J6-X	J7-D	J10-A	J8-R
J7-A	J7-B	J10-A	J8-S
J7-f	J8-b	J14-AC	J14-g
J7-H	J8-AT	J14-AG	J16-K
J7-L	J8-V	J14-c	J15-c
J7-n	J8-K	J14-k	J16-h
J7-P	J8-B	J14-M	J15-M
J7-V	J8-AV	J14-M	J16-F
J8-A	J3-F	J14-m	J16-g
J8-A	J9-K	J14-N	J16-C
J8-AA	J6-p	J14-n	J16-L
J8-AA	J6-r	J14-O	J16-P
J8-AA	J6-s	J14-P	J16-E
J8-AA	J6-T	J14-p	J16-J
J8-AA	J14-R	J14-S	J7-E
J8-AM	J4-Y	J14-T	J16-T
J8-AW	J9-A	J14-u	J16-N
J8-AX	J9-C	J14-U	J14-a
J8-AY	J9-D	J14-U	J14-b
J14-U	J14-Z	J15-u	J16-V
J14-v	J16-M	J15-U	J15-a
J14-w	J16-e	J15-U	J15-b
J14-x	J16-f	J15-U	J15-Z
J14-y	J16-B	J15-v	J16-U
J14-z	J16-A	J15-w	J15-y
J15-AA	J15-AC	J15-y	J15-AA
J15-AG	J16-c	J16-a	J14-f
J15-f	J16-W	J16-b	J14-g
J15-g	J15-n	J16-S	J14-s
J15-g	J15-X		
J15-g	J16-X		

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TABLE V. Continuity - Continued.

From connector pin	To connector pin	From connector pin	To connector pin
J15-k	J16-Y		
J15-m	J16-Z		
J15-n	J15-w		
J15-R	J16-R		
J15-T	J16-D		

3.5 Environmental conditions. The VDB shall demonstrate no performance degradation beyond specification limits and show no evidence of damage or deformation when subjected to the following environments.

3.5.1 Low temperature. The VDB shall demonstrate specified wire continuities before and after exposure to low temperatures and shall meet the performance requirements of 3.4.2.1 through 3.4.11 after storage at -70 degrees Fahrenheit (°F) (-56.7 degrees Celsius (°C)) and during exposure to temperature as low as -65°F (-54°C) (see 4.8.1).

3.5.2 High temperature. The VDB shall demonstrate specified wire continuities before and after exposure to high temperatures and shall meet the performance requirements of 3.4.2.1 through 3.4.11 after storage at temperatures as high as 160°F (71°C) and during exposure to a temperature of 130°F (54.5°C) (see 4.8.2).

3.5.3 Shock. The VDB shall meet the performance requirements of 3.4.2.1 through 3.4.11 after being subjected to sawtooth shock pulses having a peak value of 40 gravity units (g) for 11 ms time duration applied along each of the three major axes (see 4.8.3).

3.5.4 Vibration. The VDB shall meet the performance requirements of 3.4.2.1 through 3.4.11 after being subjected to sinusoidal vibration levels of 1.5 and 4.2 g in the sweep frequency range of 5 to 500 to 5 Hz with a sweep time of 15 minutes (see 4.8.4).

3.5.5 Steam and waterjet cleaning. The VDB shall meet the performance requirements of 3.4.2.1 through 3.4.11 following a steam and waterjet cleaning process which uses a commercial heavy-duty non-abrasive cleaner. Jet pressure shall be 105 $\pm$ 5 pounds per square inch gage (psig) (724 $\pm$ 34 kilopascals (kPa)) for steam and 50 $\pm$ 5 psig (345 $\pm$ 34 kPa) for water. The jet shall be applied perpendicular to the VDB surface from a distance of not more than 1 foot (ft) (30 centimeters (cm)) for steam and not more than 3 ft (91.5 cm) for water, and moved at such a rate that the exposed surface of the VDB is subjected to the jet at the rate of 1 square foot per minute (ft^2/min) (929 square centimeters per minute (cm^2/min)) (see 4.8.5).

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3.5.6 Humidity. The VDB shall demonstrate: specified wire continuities before and after exposure to relative humidity of 94 + 4% at 86°F (30°C) over a period of 24 hours; and shall meet the requirements of 3.4.2.1 through 3.4.11 after exposure to this humid environment (see 4.8.6).

3.5.7 Fungus. The VDB shall show no evidence of fungus growth after being subjected to a fungus conducive environment. The VDB shall then meet the requirements of 3.4.2.1 through 3.4.11 at the conclusion of the fungus test (see 4.8.7).

3.6 Identification marking. The VDB shall be legibly and permanently marked and shall include as a minimum the following (see 4.7.2):

Nomenclature: Chassis Distribution Box
Design Activity: 19207
Part Number: 12328513
Manufacturer's Serial Number
Manufacturer's Cage Code
Date of Manufacturing

3.7 Workmanship. All parts and components of the VDB shall be free from dirt and other extraneous material and from defects of workmanship that impair performance (see 4.7.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Conformance inspections (CI) (see 4.5).
c. Control tests (see 4.6).

4.2 Inspection equipment.

4.2.1 Automatic test equipment (ATE). ATE shall meet minimum power, accuracy and loading requirements to test the VDB (see 6.3).

4.2.1.1 ATE software. ATE software shall be in accordance with a software plan approved by the procuring activity. This plan shall address as a minimum the following software-related activities:

a. Design and coding standards
b. Configuration management
c. Testing

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d. Documentation
e. Quality assurance, including corrective action

In cases where a circuit is common to more than one function, the program shall test the circuit one time for one function, but shall not be required to retest the circuit for the additional function(s) (see 6.4.3).

4.2.1.1.1 Data systems compatibility. Compatibility shall be established between the contractor's and the procuring activity's data systems at the time of contract award (see 6.2).

4.2.1.2 Fault insertion. Unless otherwise specified herein, ATE shall have fault insertion performed in accordance with Appendix A, whenever ATE is developed or altered (see 6.2).

4.3 Inspection conditions. Unless otherwise specified herein (see 6.2), all inspections shall be performed under the following standard (room) ambient conditions:

a. Temperature: 77 ± 18°F (25 ± 10°C).
b. Relative humidity: Uncontrolled room ambient.
c. Atmospheric pressure: Site pressure.

4.4 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on three samples as specified in table VI. Group C except the fungus test shall be performed on the first two samples while the fungus test shall be performed on the third sample only.

4.5 Conformance inspections (CI). CI shall be conducted on VDBs and shall include the examination of 4.5.2 and the tests of 4.5.3.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the VDBs from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for inspection. Samples for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4.

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TABLE VI. Classification of inspections.

Title	Requirement	Inspection	First article	Conformance		Control
				Examination	Test (100%)	
Group A:						
Materials & construction	3.2 thru 3.3.1 & 3.3.3	4.7.1	X			
Defects (see table VII)	3.3, 3.3.2, 3.6, & 3.7	4.7.2	X	X		
Group B (Performance):						
Primary power source	3.4.1	4.7.3	X		X	
Cold start relay, K1	3.4.2.1	4.7.4.1	X		X	
Fire suppressant relay, K8	3.4.2.2	4.7.4.2	X		X	
Starter lockout relay, K3	3.4.2.3 1/	4.7.4.3	X		X	
Lamp test relay, K4	3.4.2.4	4.7.4.4	X		X	
Master/slave power transfer relay, K5	3.4.2.5	4.7.4.5	X		X	
Hydraulic pump motor relay, K7	3.4.2.6	4.7.4.6	X		X	
NBC system operation	3.4.3	4.7.5	X		X	
Warning signal functions	3.4.4 2/	4.7.6	X		X	
Headphone warning signal	3.4.4.1 3/	4.7.6	X		X	
Flashing indicator signal	3.4.4.2	4.7.6	X		X	
Vehicle motion switching function	3.4.5 3/	4.7.7	X		X	
Resistance measurements	3.4.6.1	4.7.8.1	X		X	
Direct-current voltage	3.4.6.2	4.7.8.2	X		X	
Top dead center (TDC) signal	3.4.6.3 3/	4.7.8.3	X		X	
Coolant low-level sending circuit	3.4.7	4.7.9	X		X	
Master power relay current-limiting function	3.4.8	4.7.10	X		X	
Lamp test	3.4.9.1	4.7.11.1	X		X	
Diode isolation	3.4.9.2	4.7.11.2	X		X	
Lamp test circuit	3.4.9.3	4.7.11.3	X		X	
Ramp control function	3.4.10	4.7.12	X		X	
Wiring continuity	3.4.11	4.7.13	X		X	
Group C (Environmental):						
Low temperature	3.5.1	4.8.1	X			X
High temperature	3.5.2	4.8.2	X			X
Shock	3.5.3	4.8.3	X			

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TABLE VI. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Conformance		Control
				Examination	Test (100%)	
Vibration	3.5.4	4.8.4	X			X
Steam & waterjet cleaning	3.5.5	4.8.5	X			
Humidity	3.5.6	4.8.6	X			X
Fungus	3.5.7	4.8.7	X			

1/ The performance requirement in 3.4.2.3 does not apply at supply voltages below 28 Vdc.
2/ Equivalent supply voltage/load resistor values may be substituted, provided that the current provided E/R remains the same (± 20%) of that indicated.
3/ The load resistors specified shall be used.

4.5.2 Examination. VDBs selected shall be examined for the defects specified in table VII.

TABLE VII. Classification of defects.

Category	Defect	Method of inspection
Major		
101	Assembly, incomplete (see 3.3).	Visual
102	Dissimilar metals not as specified (see 3.2.1).	Visual
Minor		
201	Lockwire, not as specified (see 3.3.2).	Visual
202	Identification marking, improper (see 3.6).	Visual
203	Faulty workmanship affecting appearance (see 3.7).	Visual

4.5.3 CI tests (100%). VDBs shall be subjected to the CI tests specified in table VI.

4.6 Control tests. Unless otherwise specified (see 6.2), control test shall be conducted on one from each 300 VDBs consecutively produced. The frequency of test shall be not more than one test performed in a 3-month period, nor less than one test in a 6-month period. VDBs selected for control tests shall be new and inspected in accordance with 4.5 prior to subjecting the sample items to the control tests specified in table VI.

4.7 Methods of inspection. The following methods of inspection are acceptable as satisfying the requirements of 3.4:

a. Direct Support Electrical Systems Test Set (DSESTS) (see 6.3).
b. The Symbolic State Sequence functional tests specified in 4.7.3 through 4.7.13, intended for use with ATE (see Appendix B).

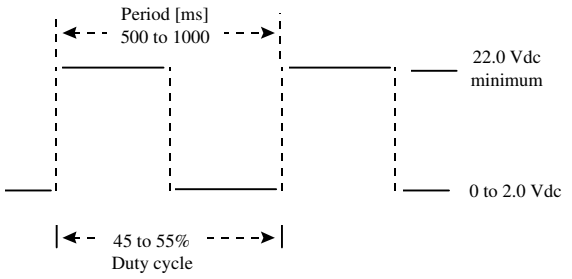
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4.7.1 Materials and construction. Conformance to 3.2 through 3.3.1 and 3.3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, receiving inspection records, processing and quality control standards, industry standards, and test reports.

4.7.2 Defects. Conformance to 3.3, 3.3.2, 3.6 and 3.7 shall be determined by examination for the defects listed in table VII. Examination shall be visual.

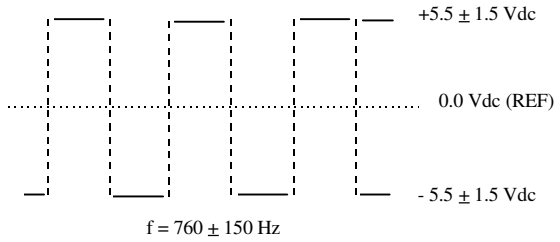
4.7.3 Set-up and definitions. To determine conformance to 3.4.1, definitions of terms and voltages shall be in accordance with table VIII. Loads shall be connected per table IX. Test interface conditions shall be in accordance with table X. See appendix B for symbols used in tables X through XXIII.

TABLE VIII. Definitions.

Term	Definition
V_0	0.0 to 0.5 Vdc
V_1	1.4 to 2.8 Vdc
V_{1a}	0.69 to 1.36 Vdc
V_2	0.0 to 2.0 Vdc
V_{10}	10.0 ± 1.0 Vdc
V_{10a}	> 10.5 but < 26.0 Vdc
V_{12}	5.0 to 12.0 Vdc
V_{12a}	6 to 20 ms pulse, amplitude within 1.5 Vdc of V_{12}, one pulse for each negative-going transition of waveform at J16-G (see 3.4.6.3).
V_{24}	18.0 to 26.0 Vdc
V_{28}	28.0 ± 0.5 Vdc
V_{28a}	28.0 ± 1.0 Vdc
V_{28b}	28.0 to 30.0 Vdc
V_{28c}	23 to 26 Vdc (source voltage minus 2.0 to 5.0 Vdc)
V_{28d}	28.0 ± 1.5 Vdc
F_{20}	2.0 ± 0.5 Vpp, 10.5 ± 9.5 Hz, Square wave.
F_{22}	

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TABLE VIII. Definitions - Continued.

Term	Definition
F_{40a}	< 2.6 V rms, 40.0 ± 2.0 Hz, sine wave.
F_{40b}	9.2 ± 0.1 V rms, 40.0 ± 2.0 Hz sine wave.
F_{575}	2.5 ± 0.5 Vpp, < 575 Hz, sine wave, source impedance > 100 ohms, J7-N reference (gnd).
F_{760}	 Signal is interrupted (output is 0 to 0.5 Vdc) at a rate of 1.5 ± 0.5 Hz.
F_{1100}	2.5 ± 0.5 Vpp, > 1100 Hz, sine wave, source impedance > 100 ohms, J7-N reference (gnd).
E_1	Ground reference for all power supplies unless otherwise specified.

NOTE: First article testing shall be accomplished with the 28 Vdc supplies adjusted and tested at high (29.5 to 30.0 Vdc) and low (18.0 to 18.5 Vdc) voltage limits in addition to 28.0 ± 0.5 Vdc.

TABLE IX. Load connections.

Condition	Load	Value
Load connected between pin J8-Z and E1 (GND)	R_1	55 ohm, 1%, 25 watt
Load connected between pin J8-Z and E1 (GND)	R_2	1 ohm, 1%, 5 watt
Load connected between pin J7-M and J7-N (GND)	R_3	100 to 150 ohm, 1/2 watt
Load connected between pin J4-K and E1 (GND)	R_4	1800 ohm, 5%, 1/2 watt
Load connected between pin J8-C and E1 (GND)	R_5	185 Kohm, 10%, 1/4 watt
Load connected between pin J14-c and V_{12}	R_6	5.6 Kohm, 5%, 1/4 watt
Load connected between pin J8-X and V_{28}	R_7	1000 to 4700 ohm, 1 watt
Load connected between pin J8-BR and E1 (GND)	R_8	1 Kohm, 5%, 1/4 watt

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TABLE X. Test interface conditions (TIC).

I.D.	Signal name	Accept LO	Accept HI	Open/ ground	Load	In- put	Out- put	Init
E1	Power GND			G				G
J1-C	Fire suppressant	V_0	V_{10}			X		LO
J1-D	NBC system	V_0	V_{28}				*	LO
J1-E	NBC system	V_0	V_{28}				*	LO
J1-F	NBC system	V_0	V_{28}				*	LO
J1-G	NBC system	V_0	V_{28}				*	LO
J2-H	Master power		V_{28}	o		X		o
J2-J	Fire suppressant	V_0	V_{28}	o		X		o
J3-C	Pump motor relay		V_{28}	o		X		o
J4-X	Launcher up lamp			o/G		X		o
			V_1				*	
J4-K	Vehicle motion sig.	V_2	V_{10a}	o	R_4		*	o
J4-W	+24 Vdc	V_o	V_{24}	o		X		o
J7-a	Air cleaner rest.			o/G		X		o
			V_1				*	
J7-c	Fuel filter rest.			o/G		X		o
			V_1				*	
J7-h	Warning sig GND			o/G		X		o
J7-A	Cold start relay	V_0	V_{28}				*	LO
J7-B	Cold start relay	V_0	V_{28}				*	LO
J7-k	Coolant low level	V_0	V_{28d}	o			*	o
J7-M	Starter lockout		F_{1100}	o/G	R_3	X		o
J7-N	Signal reference		F_{575}	o/G	R_3	X		o
J7-R	Engine oil pressure			o/G		X		o
			V_1				*	
J7-S	Coolant high temp			o/G		X		o
			V_1				*	
J7-T	Trans oil temp			o/G		X		o
			V_1				*	
J7-U	Trans oil press			o/G		X		o
			V_1				*	
J8-C	Headphone signal	V_0	F_{760}		R_5		*	LO
J8-E	Cold start relay			o/G		X		o
J8-U	Pump motor relay		V_{28}	o		X		o
J8-X	Coolant low level		V_{28}	o	R_7	X		o
J8-Z	Master pwr relay	V_{1a}	V_{28c}		R_1, R_2		*	LO
J8-AA	28 Vdc power bus		V_{28}	o		X		o
			V_{28b}	o		X		
J8-AC	Tone cancel			o/G		X		o
J8-AF	Lamp test		V_{28}	o		X		o
J8-AJ	Lamp flasher bus	V_0	F_{22}				*	LO

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TABLE X. Test interface conditions (TIC) - Continued.

I.D.	Signal name	Accept LO	Accept HI	Open/ ground	Load	In- put	Out- put	Init
J8-BC	Starter lockout		E1	o			*	o
J8-BD	Pump motor GND			o/G		X		G
J8-BR	Discharge lamp	V_0	V_{28}		R_8		*	LO
J9-M	F40 reference (gnd)			o/G		X		o
J9-N	Veh. motion sig.	F_{40a}	F_{40b}	o		X		o
J10-A	Veh. main bus		V_{28}	o		X		o
			V_{28a}	o			*	
J13(+)	J13 (inner socket)		V_{28}	o/G		X		o
			V_{28}	o			*	o
J13(-)	J13 (outer shell)		V_{28}	o/G		X		o
J14-c	Engine TDC signal		V_{12}		R_6	X		o
J15-c	TDC signal	V_0	V_{12a}				*	LO
J15-L	Starter lockout		E1	o			*	o
J15-V	Pump motor relay		J10-A	o			*	o
J16-G	Engine TDC	V_0	F_{20}			X		LO
J17-A	Pump motor relay		J10-A	o			*	o
	NBC system switch							OFF

4.7.4 Relay switching functions.

4.7.4.1 Cold start relay, K1. To determine conformance to 3.4.2.1, the cold start relay shall be tested in accordance with table XI.

TABLE XI. Cold start relay.

Seq. No.	Operation	Remarks
101	Begin	3.4.2.1
102	Init:	
103	J10-A/HI & J9-E/G:	
104	*J7-A/HI & *J7-B/HI:	
105	End	

4.7.4.2 Fire suppressant relay, K8. To determine conformance to 3.4.2.2, the fire suppressant relay shall be tested in accordance with table XII.

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TABLE XII. Fire suppressant relay.

Seq. No.	Operation	Remarks
201	Begin	3.4.2.2
202	Init:	
203	J2-J/HI:	
204	# Activate switch S2:	
205	*J8-BR/LO:	3.4.2.2a
206	J1-C/#HI (75 < P < 175):	
207	*J8-BR/HI:	3.4.2.2b
208	End	

4.7.4.3 Starter lockout relay, K3. To determine conformance to 3.4.2.3, the starter lockout relay shall be tested in accordance with table XIII.

TABLE XIII. Starter lockout relay.

Seq. No.	Operation	Remarks
301	Begin	3.4.2.3
302	Init:	This test not performed at low voltage (source 18 Vdc)
303	J8-AA/HIb:	
304	J7-M/G:	
305	*J8-BC/HI & *J15-L/HI:	3.4.2.3a verify continuity
306	J7-M/F_{575}:	
307	*J8-BC/HI & *J15-L/HI:	3.4.2.3b verify continuity
308	J7-M/F_{1100}:	
309	*J8-BC/o & *J15-L/o:	3.4.2.2b
310	End	

4.7.4.4 Lamp test relay, K4. To determine conformance to 3.4.2.4, the lamp test relay shall be tested in accordance with table XIV.

TABLE XIV. Lamp test relay.

Seq. No.	Operation	Remarks
401	Begin	3.4.2.4, 3.4.4c
402	Init:	
403	J8-AA/HI:	
404	J8-AF/HI & J10-A/HI:	
405	*J8-C/F_{760} & *J8-AJ/F_{22}:	3.4.2.4a, 3.4.4.1 3.4.2.4b, 3.4.4.2
406	End	

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4.7.4.5 Master/slave power transfer relay, K5. To determine conformance to 3.4.2.5, the master/slave power transfer relay shall be tested in accordance with table XV.

TABLE XV. Master/slave power transfer relay.

Seq. No.	Operation	Remarks
501	Begin	3.4.2.5.1
502	Init:	
503	J13(-)/G & J13(+)/HI:	J13 (outer shell) J13 (inner socket)
504	*J10-A/HIa:	3.4.2.5.1a
505	*Verify "slave receptacle power" lamp green portion on:	3.4.2.5.1b, lamp green
506	End	
507	Begin	3.4.2.5.2
508	Init:	
509	J13(+)/G & J13(-)/HI:	J13 (inner socket) J13 (outer shell)
510	*J10-A/Lo:	3.4.2.5.2a
511	*Verify "slave receptacle power" lamp red portion on:	3.4.2.5.2b, lamp red
512	End	
513	Begin	3.4.2.5.3
514	Init:	
515	J10-A/HI:	
516	Depress "energize slave receptacle" pushbutton and hold: *J13(+)/HI:	Manual operation, 3.4.2.5.3
517	End	

4.7.4.6 Hydraulic pump motor relay, K7. To determine conformance to 3.4.2.6, the hydraulic pump motor relay shall be in accordance with table XVI.

TABLE XVI. Hydraulic pump motor relay.

Seq. No.	Operation	Remarks
601	Begin	3.4.2.6
602	Init:	
603	J8-BD/G & J8-U/HI:	
604	*J17-A/HI & *J15-V/HI:	
605	Init:	
606	J8-BD/G & J3-C/HI:	
607	J/604	
608	End	

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4.7.5 NBC system operation. To determine conformance to 3.4.3, the NBC system operation shall be tested in accordance with table XVII.

TABLE XVII. NBC system operation.

Seq. No.	Operation	Remarks
701	Begin	3.4.3
702	Init:	
703	J10-A/HI:	
704	Place "NBC system" switch to "ON":	Manual operation
705	*J1-E/HI & *J1-F/HI & *J1-G/HI & *J1-D/HI:	
706	*Verify "NBC system" lamp is "ON"	Visual operation
707	End	

4.7.6 Warning signal functions. To determine conformance to 3.4.4, the warning signal functions (headphone warning/flashing indicator) shall be tested in accordance with table XVIII.

TABLE XVIII. Warning signal functions.

Seq. No.	Operation	Remarks
801	Begin	3.4.4
802	Init:	
803	J8-AA/HI:	
804	J4-X/G:	3.4.4a
805	*J8-C/F_{760} & *J8-AJ/F_{22}:	3.4.4.1, 3.4.4.2
806	J8-AC/G:	
807	*J8-C/LO & *J8-AJ/Lo:	
808	J4-X/o & J8-AC/o:	
809	J7-R/G:	3.4.4a
810	*J8-C/F_{760} & *J8-AJ/F_{22}:	3.4.4.1, 3.4.4.2
811	J8-AC/G:	
812	*J8-C/LO & *J8-AJ/Lo:	
813	J7-R/o & J8-AC/o:	
814	J7-S/G:	3.4.4a
815	*J8-C/F_{760} & *J8-AJ/F_{22}:	3.4.4.1, 3.4.4.2
816	J8-AC/G:	
817	*J8-C/LO & *J8-AJ/Lo:	
818	J7-S/o & J8-AC/o:	
819	J7-T/G:	3.4.4a
820	*J8-C/F_{760} & *J8-AJ/F_{22}:	3.4.4.1, 3.4.4.2
821	J8-AC/G:	

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TABLE XVIII. Warning signal functions - Continued.

Seq. No.	Operation	Remarks
822	*J8-C/LO & *J8-AJ/Lo:	
823	J7-T/o & J8-AC/o:	
824	J7-U/G:	3.4.4a
825	*J8-C/F_{760} & *J8-AJ/F_{22}:	3.4.4.1, 3.4.4.2
826	J8-AC/G:	
827	*J8-C/LO & *J8-AJ/Lo:	
828	J7-U/o & J8-AC/o:	
829	J7-a/G:	3.4.4a
830	*J8-C/F_{760} & *J8-AJ/F_{22}:	3.4.4.1, 3.4.4.2
831	J8-AC/G:	
832	*J8-C/LO & *J8-AJ/Lo:	
833	J7-a/o & J8-AC/o:	
834	J7-c/G:	3.4.4a
835	*J8-C/F_{760} & *J8-AJ/F_{22}:	3.4.4.1, 3.4.4.2
836	J8-AC/G:	
837	*J8-C/LO & *J8-AJ/Lo:	
838	J7-c/o & J8-AC/o:	
839	J8-X/HI & J7-h/G:	3.4.4b
840	*J8-C/F_{760} & *J8-AJ/F_{22}:	3.4.4.1, 3.4.4.2
841	J8-AC/G:	
842	*J8-C/LO & *J8-AJ/Lo:	
843	J8-X/o & J8-AC/o:	
844	J8-AF/HI & J10-A/HI:	3.4.4c
845	*J8-C/F_{760} & *J8-AJ/F_{22}:	3.4.4.1, 3.4.4.2
846	J8-AC/G:	
847	*J8-C/LO & *J8-AJ/Lo:	
848	End	

4.7.7 Vehicle motion switching function. To determine conformance to 3.4.5, the vehicle motion switching function shall be tested in accordance with table XIX.

TABLE XIX. Vehicle motion switching function.

Seq. No.	Operation	Remarks
901	Begin	3.4.5
902	Init:	
903	J4-W/HI & J4-K/E1 (R4) & J9-M/G & J9-N/F_{40a}:	J4-K connected to E1 thru R4
904	*J4-K/LO:	3.4.5a
905	J9-N/F_{40b}:	
906	*J4-K/HI:	3.4.5b
907	End	

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4.7.8 STE M1/FVS and TMDE functions.

4.7.8.1 Resistance measurements. To determine conformance to 3.4.6.1, the resistance measurement, shall be tested in accordance with table I.

4.7.8.2 Direct current voltages. To determine conformance to 3.4.6.2, the direct-current voltage shall be tested in accordance with table II. A voltage source shall be applied between pins J16-a(+) and J16-b(-) and pins J16-W(+) and J16-X(-). The voltage source shall be greater than 5.0 Vdc but shall not exceed 40.0 Vdc. The voltage measured between the pin pairs shall be the corresponding fraction of the supply voltage as stated in table II.

4.7.8.3 Top dead center (TDC) signal. To determine conformance to 3.4.6.3, the TDC shall be tested in accordance with table XX.

TABLE XX. TDC signal.

Seq. No.	Operation	Remarks
1001	Begin	3.4.6.3
1002	Init:	
1003	J8-AA/HI:	3.4.6.3a
1004	J14-c/HI:	3.4.6.3b
1005	J16-G/F_{20}:	3.4.6.3c
1006	*J15-c/HI:	figure 3
1007	End	

4.7.9 Coolant low-level sending circuit. To determine conformance to 3.4.7, the coolant low-level sending circuit tests shall be in accordance with table XXI.

TABLE XXI. Coolant low-level sending circuit.

Seq. No.	Operation	Remarks
1101	Begin	3.4.7.1
1102	Init:	
1103	J8-AA/HI:	
1104	*J7-k/HI:	
1105	End	

4.7.10 Master power relay current-limiting function. To determine conformance to 3.4.8, the master power relay current-limiting function tests shall be in accordance with table XXII.

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TABLE XXII. Master power relay current-limiting function.

Seq. No.	Operation	Remarks
1301	Begin	3.4.8
1302	Init:	
1303	J8-Z/E1 (R1) & J10-A/HI:	Connect J8-Z to E1 thru R1, 450 ± 50 mA load.
1304	*J8-Z/HI:	
1305	J10-A/o & J2-H/HI:	
1306	*J8-Z/HI & J10-A/o:	
1307	J8-Z/E1 (R2) & J10-A/HI:	Connect J8-Z to E1 thru R2, verify current limiting to 0.70 to 1.35 amps.
1308	*J8-Z/LO:	
1309	J10-A/o & J2-H/HI:	
1310	*J8-Z/LO:	
1311	End	

4.7.11 Indicator lamp control functions.

4.7.11.1 Lamp test. To determine conformance to 3.4.9.1, the lamp test function shall be tested in accordance with table XXIII.

TABLE XXIII. Indicator lamp control functions.

Seq. No.	Operation	Remarks
1401	Begin	3.4.9.1
1402	Init:	
1403	J8-AA/HI & J8-AF/HI:	
1404	*J7-U/HI & *J7-S/HI & *J7-X/HI & *J7-a/HI & *J7-c/HI & *J7-T/HI & *J7-R/HI:	
1405	End	

4.7.11.2 Diode isolation. To determine conformance to 3.4.9.2, the diode isolation as specified in table III shall be verified.

4.7.11.3 Lamp test circuits. To determine conformance to 3.4.9.3, following the application of 28.0 ± 0.5 Vdc to J8-AF, the lamp test circuitry as specified in table IV shall be verified.

NOTE: Internal series resistance varies in each of the circuits; external loading/current-limiting shall be provided to limit current flow in any single line to no more than 110 mA.

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4.7.12 Ramp control function. To determine conformance to 3.4.10, diode isolation shall be verified to exist between J8-BD (anode) and J8-BT (cathode).

4.7.13 Wiring continuity. To determine conformance to 3.4.11, continuity between the points specified in table V shall be verified.

4.8 Environmental conditions.

4.8.1 Low temperature. To determine conformance to 3.5.1, the VDB shall be tested as follows:

a. Install the VDB in the test chamber in a manner simulating service usage. At ambient room condition (see 4.3), the VDB shall be tested for the following wire continuities:

J2-B to J14-X
J2-C to J14-W
J2-C to J14-Y
J2-C to J15-W
J2-J to J2-K

b. The internal chamber temperature shall be lowered to a storage temperature of -70°F (-57°C) and maintained for a period of 24 hours after temperature stabilization.
c. The internal chamber temperature shall be adjusted to -65°F (-54°C), the lowest operating temperature the VDB is designed to operate. The -65°F (-54°C) environment shall be maintained until temperature stabilization is reached.
d. When stabilization at the lowest operating temperature, -65°F (-54°C), is maintained, the VDB shall be subjected to the tests specified in 4.7.4.1 through 4.7.13 (except 4.7.6) at the high and low voltages specified in 3.4.1. Starter lockout relay, 4.7.4.3, shall be tested at 28.0 and 30.0 Vdc.
e. The VDB shall be returned to nonoperating ambient room conditions (see 4.3) and allowed to stabilize.
f. The VDB shall be inspected for wire continuity as specified in step a.

4.8.2 High temperature. To determine conformance to 3.5.2, the VDB shall be tested as follows:

a. Install the VDB in the test chamber in a manner simulating service usage. At ambient room condition (see 4.3) the VDB shall be tested for the following wire continuities:

J2-B to J14-X
J2-C to J14-W

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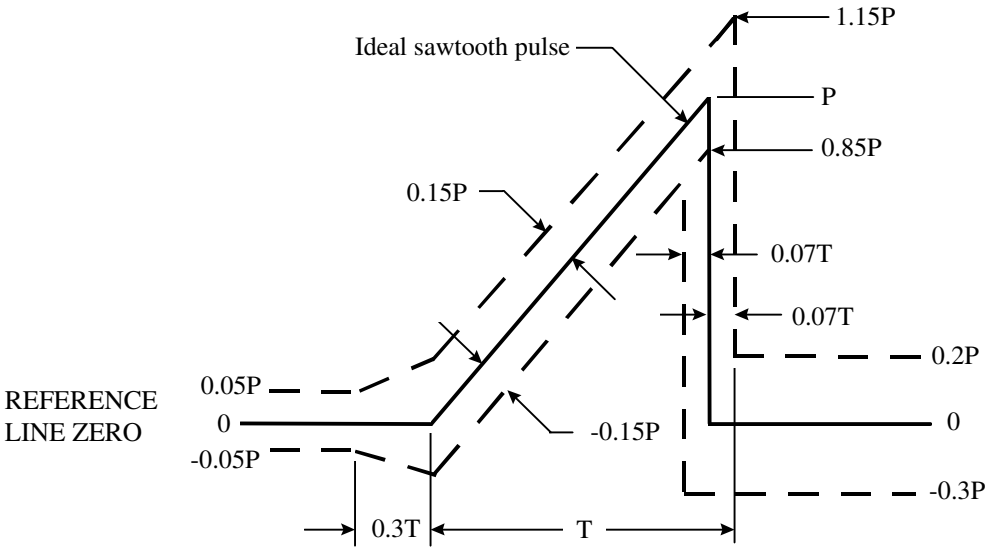
J2-C to J14-Y
J2-C to J15-W
J2-J to J2-K

b. The internal chamber temperature shall be raised to 160°F (71°C) and maintained for 48 hours while ensuring relative humidity is not greater than 15%.
c. The internal chamber temperature shall be adjusted to 130°F (54.5°C), the highest operating temperature at which the VDB is designed to operate. The 130°F (54.5°C) environment shall be maintained until temperature stabilization is reached.
d. When stabilization at the highest operating temperature, 130°F (54.5°C), is maintained, the VDB shall be subjected to the tests specified in 4.7.4.1 through 4.7.13 (except 4.7.6) at the high and low voltages specified in 3.4.1. Starter lockout relay, 4.7.4.3, shall be tested at 28.0 and 30.0 Vdc.
e. The VDB shall be returned to nonoperating ambient room conditions (see 4.3) and allowed to stabilize.
f. The VDB shall be inspected for wire continuities as specified in step a.

4.8.3 Shock. To determine conformance to 3.5.3, the VDB shall be rigidly mounted to a shock table and three shocks in each direction shall be applied along three mutually perpendicular axes (total of 18 shocks) while the VDB is nonoperational. The pulse shape shall be in accordance with figure 4. At the conclusion of the test, the VDB shall be subjected to the tests specified in 4.7.4.1 through 4.7.13.

4.8.4 Vibration. To determine conformance to 3.5.4, the VDB shall be rigidly mounted to a vibration table and subjected to sinusoidal vibration along each axis in accordance with figure 5. The sweep time shall be 15 minutes for the sweep frequency range of 5 to 500 to 5 Hz. When resonances below 5 Hz are measured, the test curves shall be extended to 2 Hz and the sweep time shall be 18 minutes (2 to 500 to 2 Hz). The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 6. The specified sweep time is that of an ascending plus a descending sweep and is twice the ascending time shown in figure 6 for the specified range. The sinusoidal cycling time shall be 30 minutes per axis. After vibration testing, the VDB shall be subjected to the tests specified in 4.7.4.1 through 4.7.13.

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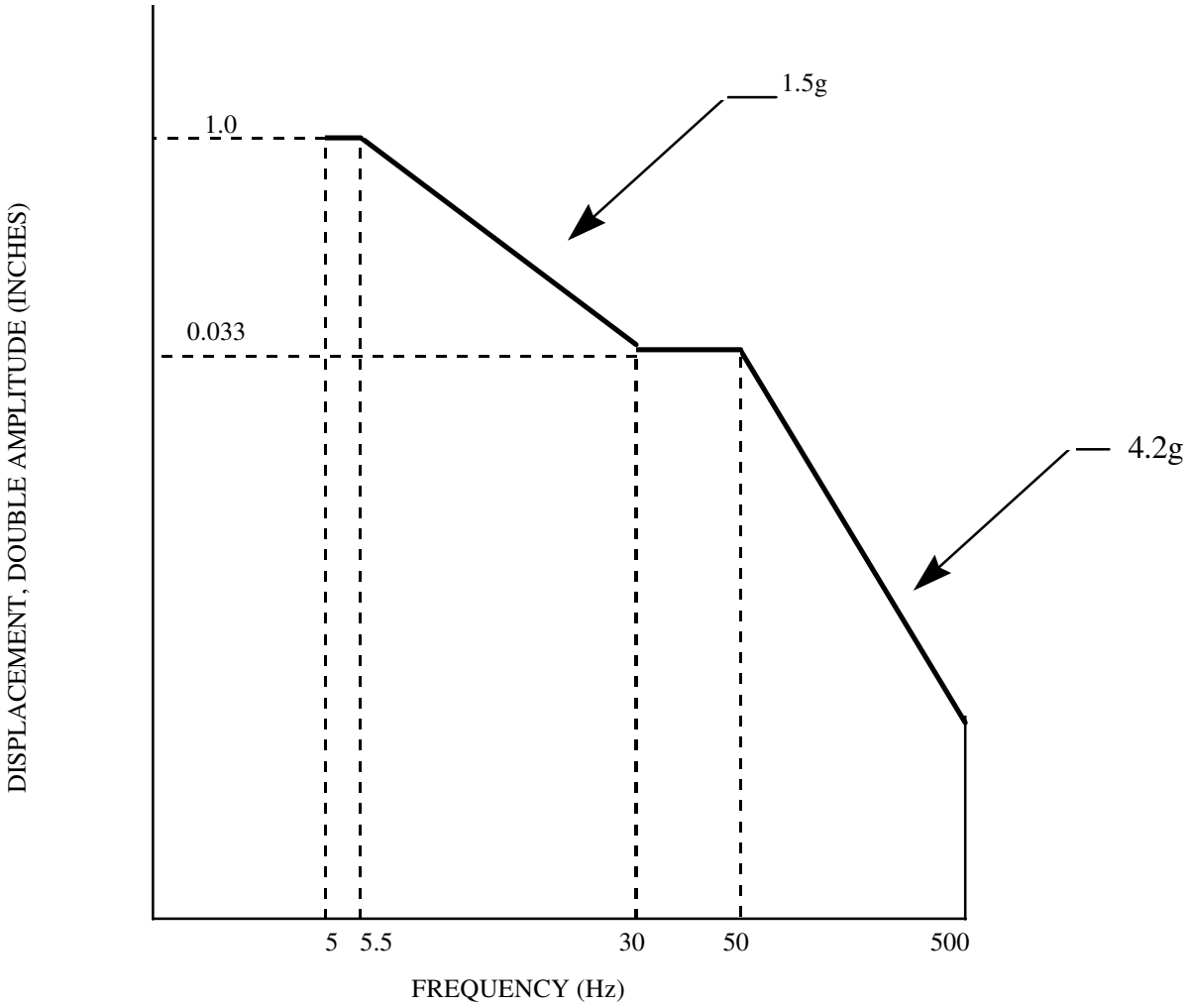


NOTES:

1. Peak value (P) = 40 g.
2. Nominal duration (T) = 11 ms.
3. The oscillogram shall include a time about 3T long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is T. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of Vi $\pm$ 0.1 Vi, where Vi is the velocity change associated with the ideal pulse which equals 0.5 PT. The integration to determine velocity change shall extend from 0.4T before the pulse to 0.1T after the pulse.

FIGURE 4. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

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NOTES:

1. Acceleration levels: ± g.
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 5. Vibration test curves for equipment installed in ground vehicles.

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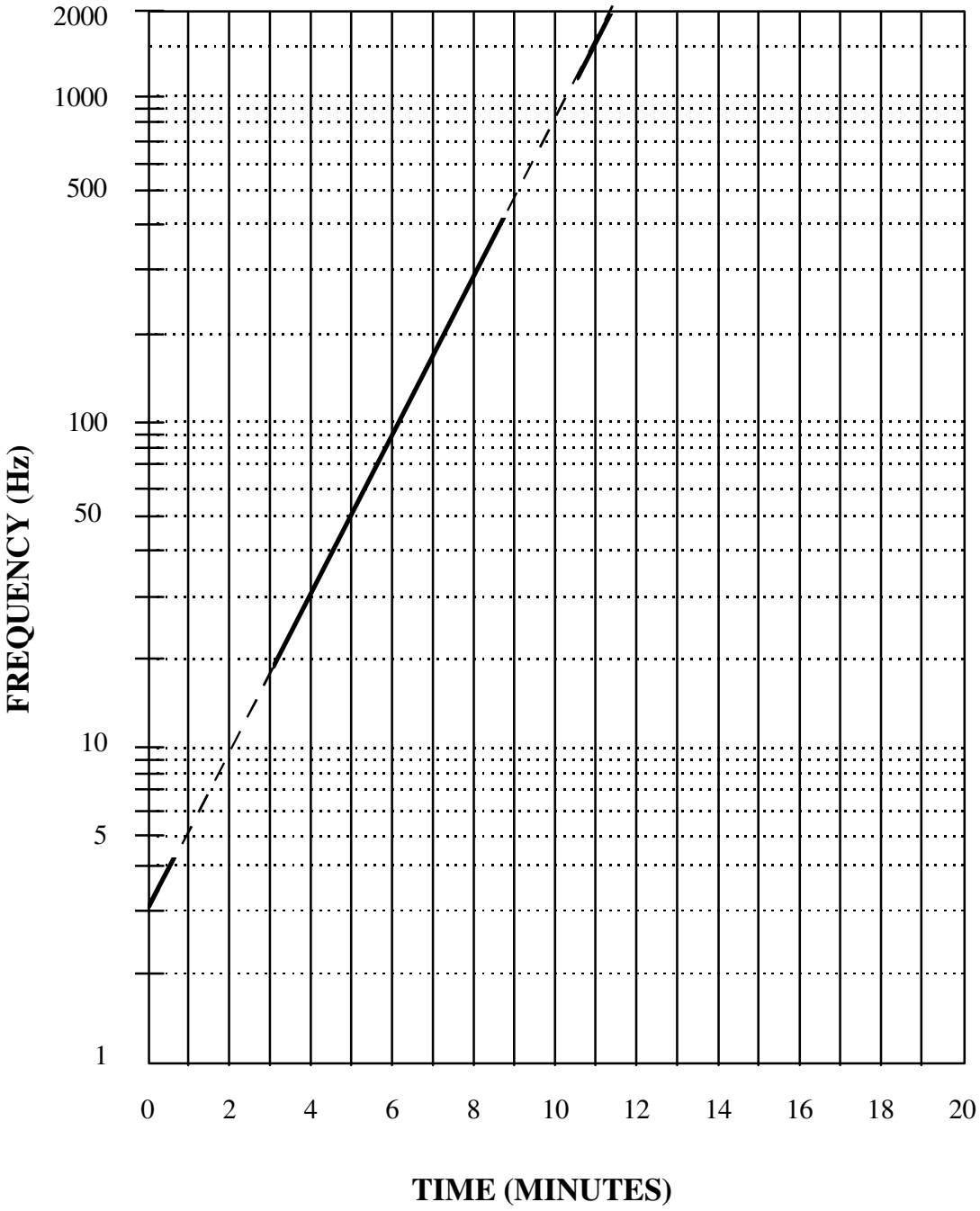


FIGURE 6. Logarithmic sweep.

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4.8.5 Steam and waterjet cleaning. To determine conformance to 3.5.5, the VDB shall be subjected to normal steam cleaning with receptacles suitable sealed. The steam and waterjet cleaning shall not be directed (within 3 inches (7.6 cm)) at drain holes. Immediately after steam cleaning, the VDB shall be subjected to normal waterjet cleaning. Within 20 minutes after completing steam and waterjet cleaning, the VDB shall be subjected to the tests specified in 4.7.4.1 through 4.7.13. Moisture inside the VDB shall not be considered a failure.

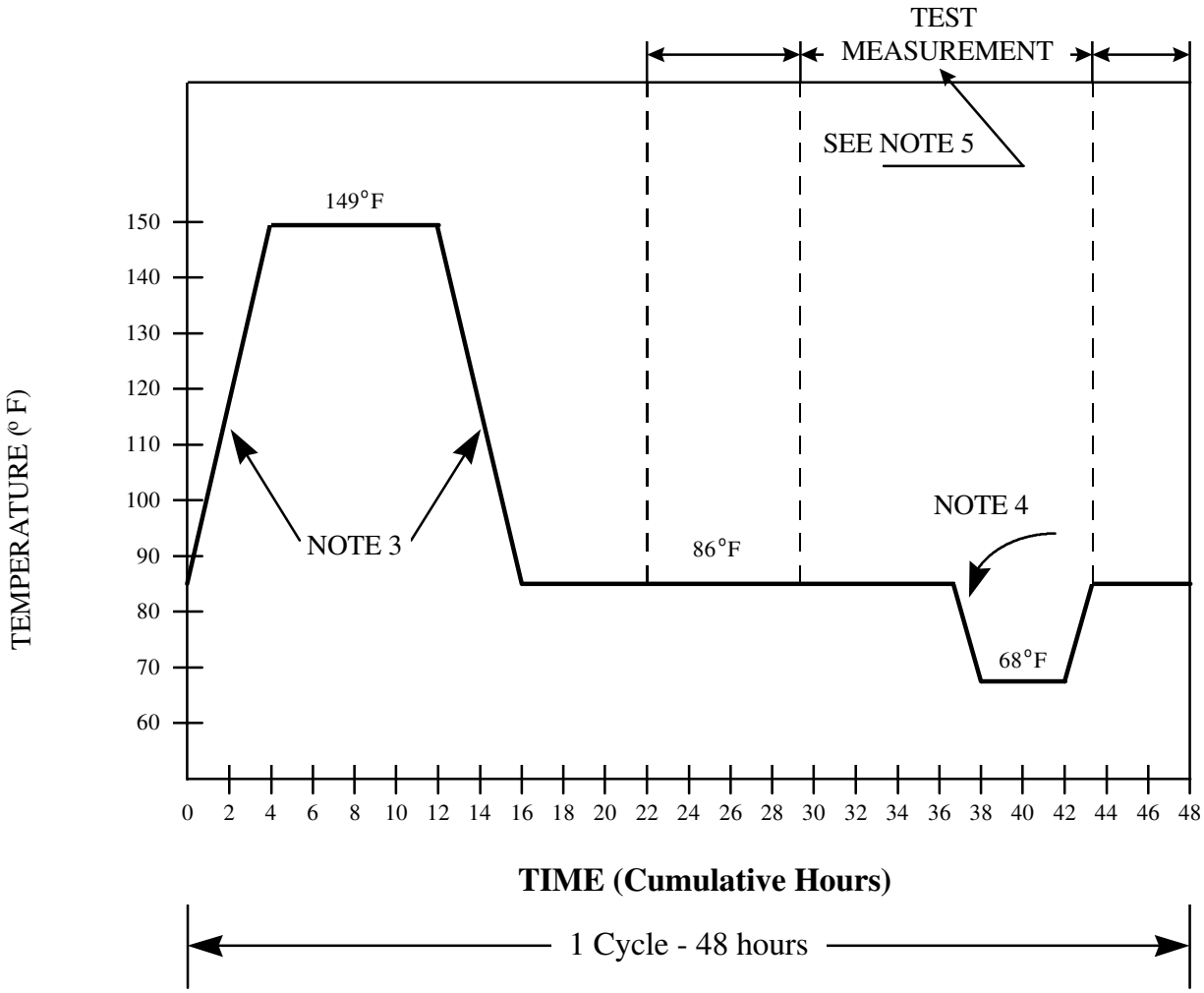
4.8.6 Humidity. To determine conformance to 3.5.6, the VDB shall be tested as follows:

a. The VDB shall be installed as in use in a chamber trap-vented to the atmosphere to prevent the buildup of total pressure. No condensation shall be permitted to drip on the VDB. Relative humidity shall be determined from dry bulb-wet bulb thermometer comparison method or an equivalent method approved by the procuring activity. Air velocity flowing across the wet bulb shall be not less than 900 feet per minute (ft/min) (274 meters per minute (m/min)). Provision shall be made for controlling the flow of air throughout the internal chamber test space where the velocity of air shall not exceed 150 ft/min. Steam or distilled, demineralized, or deionized water having a hydrogen ion concentration (pH) between 6.0 and 7.2 at 73°F (22.8°C) shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the VDB by the test facility.
b. At ambient room condition (see 4.3), the VDB shall be tested for the following wire continuities:

J2-B to J14-X
J2-C to J14-W
J2-C to J14-Y
J2-C to J15-W
J2-J to J2-K

c. After having been dried for 24 hours at 129°F (54°C), the VDB shall be conditioned at 73°F (22.8°C) and 50 $\pm$ 10% relative humidity for 24 hours. Raise the internal chamber temperature to 86°F and relative humidity to 94 $\pm$ 4%. Expose the VDB to five continuous 48-hour cycles in accordance with figure 7.
d. The VDB shall be subjected to functional tests during the fifth cycle. Prior to testing, accumulated moisture may be removed by turning the VDB upside down or shaking. Wiping is not permitted.
e. After completion of cycling, condition the VDB for 24 hours at 73°F (22.8°C) and 50 $\pm$ 10% relative humidity. At the conclusion of the humidity test the VDB shall be subjected to the tests specified in 4.7.4.1 through 4.7.13 and shall subsequently be examined for wire continuities specified in step b. Moisture inside the VDB shall not be considered a failure.

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NOTES:

1. Tolerance during temperature change shall be not greater than 5°F (2.8°C).
2. Relative humidity shall be maintained at 94 ± 4% at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85%.
3. Rate of temperature change between 86 and 149°F (30 and 65°C) shall be not less than 14.4°F (8°C) per hour.
4. The temperature increase in this portion of the curve shall be not less than 18°F (10°C).
5. Test measurements shall be taken only during the fifth humidity cycle.

FIGURE 7. Humidity cycle.

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4.8.7 Fungus. To determine conformance to 3.5.7, the VDB shall be subjected to a fungus conducive environment when specified (see 6.2), in accordance with MIL-STD-810, method 508.3. The test time shall be 28 days, no special test item configuration instructions are needed for this test. No operational tests shall be performed during the fungus test. At the conclusion of the fungus test, the VDB shall be subjected to the tests specified in 4.7.4.1 through 4.7.13. Fungus testing may be waived providing that the supplier produce certification of material not conducive to fungus growth to the satisfaction of the contracting authority.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, of by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. VDBs manufactured under this specification are intended for use on military tracked vehicles in the Bradley Fighting Vehicle Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.3).
c. If first article is required (see 3.1 and 4.4).
d. If ESS is not required (see 3.3.3).
e. Approved data system compatibility and fault insertion (see 4.2.1.1.1 and 4.2.1.2).
f. If fault insertion is not required (see 4.2.1.2).
g. If inspection conditions other than specified (see 4.3).
h. Types of first article samples, and number of samples to be inspected if other than specified (see 4.4).
i. Control test sample size and frequency of tests if other than specified (see 4.6).
j. If fungus testing should not be required (see 4.8.7).
k. Packaging requirements (see 5.1).

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6.3 Direct Support Electrical System Test Set (DSESTS) ATE. An augmented DSESTS utilizing an extended VDB Test Program Set (TPS) as specified in Drawing 4222806 is available for automatic testing (see 4.1.1).

6.4 Definitions.

6.4.1 Item definition. The VDB performs the following functions:

a. Relay switching which consist of the following:

1. Cold start relay, K1.
2. Fire suppressant relay, K8.
3. Starter cutout relay, K3.
4. Lamp test relay, K4.
5. Master/slave power transfer relay, K5.
6. Hydraulic pump motor relay, K7.

b. Gas particulate filter (GPFU) operation.
c. Warning signal generation.
d. Vehicle motion switching.
e. Connection of extra-vehicle test equipment-Simplified Test Equipment/Internal Combustion Engine (STE M1/ICE) and Test Measurement Diagnostic Equipment (TMDE).
f. Coolant low-level sending circuit.
g. Master power relay current-limiting function.
h. Indicator lamp control.
i. Ramp control.
j. Distribution of vehicle power.

6.4.2 Environmental stress screening (ESS). ESS is a process that subjects devices to physical and climatic stress (or combinations thereof). This stress forces flaws which are not ordinarily apparent into observable failures which can then be removed. Flaws are defects in design, workmanship, or material that will eventually result in the failure of the particular item.

6.4.3 Common circuit function. A common circuit function is any circuit that shares an operation with more than one stimulus, or two points that are hard-wired to each other.

6.5 Subject term (key word) listing.

Electrical; vehicle power relay
Relay switching; lamps, indicator, starter, motor and fire suppressant
Transfer power relay

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6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

FAULT INSERTION

A.1 SCOPE

A.1.1 Scope. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

A.1.2 ATE fault capability. Table XXV contains a list of each type of fault that can be inserted into the ATE to verify its capability of detecting a faulty VDB. If the ATE is capable of printing the actual measured values, the printout of fault measurements shall be verified to be within the acceptable ATE reading range for each fault. Acceptable ATE readings allow for ATE and fault insertion equipment tolerances.

A.2 APPLICABLE DOCUMENTS. No applicable documents are cited in this appendix.

A.3 FAULT CONDITIONS

A.3.1 Fault/Acceptable ATE reading derivation. The faults and acceptable fault readings were determined based on the assumption that all measuring equipment will have a combined accuracy of 95% or better, including cable loss. The faults listed in table XXV are derived with the following equations:

Given that the specification requires that a voltage fall between “X-Vdc” and “Y-Vdc”, then:

1. Below specification requirement:

    a. Fault = (X - .12Y $\pm$ .05Y) Vdc, but no less than 0.0.

2. Above specification requirement:

    a. Fault = (Y + .12Y $\pm$ .05Y) Vdc.

Acceptable ATE readings are determined by the following:

Given a fault “Z-Vdc”, which is within the defined fault to insert range, is applied to the ATE. An acceptable ATE reading shall be:

Acceptable ATE reading = (Z $\pm$ .05Z) Vdc.

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APPENDIX A

TABLE XXV. Fault conditions.

Specification Paragraph	Requirement	Fault to Insert
3.4.2.1	within 0.5 Vdc of + 28 Vdc source	0.54 to 0.59 Vdc less than + 28 Vdc source
3.4.2.2a	0 to 0.5 Vdc	0.54 to 0.59 Vdc
3.4.2.3a	0 to 1.0 ohm	1.07 to 1.17 ohms
3.4.2.3b	> 1 megohm	0.83 to 0.93 megohm
3.4.4.1	610 to 910 Hz	455 to 546 Hz
		973 to 1065 Hz
	1.0 to 2.0 Hz	0.66 to 0.86 Hz
		2.14 to 2.34 Hz
	4.0 to 7.0 Vdc	2.81 to 3.51 Vdc
		7.49 to 8.19 Vdc
	-4.0 to -7.0 Vdc	-2.81 to -3.51 Vdc
		-7.49 to -8.19 Vdc
3.4.4.2	500 to 1000 ms	330 to 430 ms
		1070 to 1170 ms
	45 to 55% (duty cycle)	35.6 to 41.2%
		58.8 to 64.4%
	0 to 2.0 Vdc	2.14 to 2.34 Vdc
	22 Vdc minimum	18.2 to 20.5 Vdc
3.4.5.b	10.5 Vdc minimum	8.72 to 9.77 Vdc
3.4.6.1	124 to 130 ohms	102 to 115 ohms
		139 to 152 ohms
	1085 to 1115 ohms	895 to 1007 ohms
		1193 to 1305 ohms
3.4.6.2	0.1144 to 0.1272%	0.0928 to 0.1055%
		0.1361 to 0.1488%
	0.475 to 0.525%	0.385 to 0.438%
3.4.6.3	6 to 20 ms	2.6 to 4.6 ms
		21.4 to 23.4 ms
	within 1.5 Vdc of (+) Vdc source	1.6 to 1.76 Vdc less than (+) Vdc source
3.4.7	within 1.0 Vdc of + 28 Vdc source	1.07 to 1.17 Vdc less than + 28 Vdc source
3.4.8	within 2.0 to 5.0 Vdc of + 28 Vdc source	1.66 to 1.86 Vdc of + 28 Vdc source
		5.35 to 5.85 Vdc of + 28 Vdc source

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APPENDIX A

TABLE XXV. Fault conditions - Continued.

Specification Paragraph	Requirement	Fault to Insert
3.4.9.1	0.70 to 1.35 A 1.4 to 2.8 Vdc	0.47 to 0.61 A 1.44 to 1.58 A 0.924 to 1.204 Vdc 2.99 to 3.28 Vdc

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APPENDIX B

SYMBOLIC STATE SEQUENCE

B.1 SCOPE

B.1.1 Scope. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.1.2 ATE verbal description. To eliminate ambiguity inherent in verbal description, and to facilitate programming and the validation of acceptance test of ATE, the functional test requirements are presented in a symbolic state sequence (SSS), using ASCII symbology and syntax described herein.

B.2 APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

B.3 REQUIREMENTS

B.3.1 Setup and interface definitions. Setup and interface definitions are presented in tabular format. These tables define all conditions, loads, initializing, etc., necessary for functional testing. Table XXVI defines examples of voltage/current tolerances and special statements.

TABLE XXVI. Example setup and interface definitions.

Term	Definition
V_{4a}	4.5 $\pm$ 0.5 Vdc
T_9	Time, read in seconds
ARRAY A	J31-C, D, E, F, G, H, J, K
F_3	3.0 $\pm$ 1.8 V_{pp}, 1.5 $\pm$ 0.8 Hz

In table XXVI, V_{4a} is defined as 4.5 $\pm$ 0.5 Vdc; all further references to V_{4a} shall be interpreted as meaning 4.5 $\pm$ 0.5 Vdc. Any other required voltages or currents shall be defined in like manner.

All special statements, such as ARRAY A, shall be defined in this table. Time T_9 is a special condition and is defined as “read in seconds”.

B.3.2 Load connections. Table XXVII defines example load connections, showing the required loads, tolerances, and wattages.

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APPENDIX B

NOTE: Pull-up/pull-down resistors, or loads used for circuit operation, are not defined. Current requirements shall be defined; contractors may tailor resistances to meet current requirements.

TABLE XXVII. Example load connections.

Condition	Load	Value
Load connected between J1-G and pins indicated in table XXIX	R1, R2	1.5 ohm, 5%, 20 watt
As specified	R3	5K, 5%, 1/4 watt

Table XXVII specifies that load R1, listed in table XXIX, shall be connected between pins J2-A and J1-G (ground); similarly, load R2, also listed in table XXIX, shall be connected between pin J2-B and J1-G (ground). The value will always be listed as a load, and called out as Rn.

Table XXVII indicates R3, as specified, shall be connected as defined in the TIC (see table XXIX).

B.3.3 Power supplies. Table XXVIII defines power connections and adjustments, and includes a reference to requirement paragraph 3.4.1 in the specification.

TABLE XXVIII. Power supplies.

Conditions	Remarks
No. 1 power supply connected to J1-D, J1-E, and J1-F	See 3.4.1 Adjust to 28.0 $\pm$ 0.5 Vdc

In accordance with table XXVIII, power connections are to be set as defined under “Conditions”, and adjustments are to be made as defined under “Remarks”.

B.3.4 Test interface conditions. Table XXIX identifies all test interface conditions (TIC), comprising all the pins used for functionally testing a unit (LRU), possible conditions, loads (if any), input(X) or output(*), and initial conditions (INIT) at the start of a test.

TABLE XXIX. Test interface conditions (TIC).

I.D.	Signal name	Accept LO	Accept HI	Open/ ground	Load	In- put	Out- put	Init
J1-n	Source			o	R3	X		o
J2-A	Random	V_{1a}	V_{28}	o	R1		*	HI
J2-B	Reference			o, G	R2		*	o
J2-n	Source	V_o	V_5	G	R3	X		LO

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APPENDIX B

B.3.4.1 TIC terminology. The terminology used in table XXIX is explained as follows:

I.D:	The test pin number.
Signal Name:	The entry "Signal Name" shall be the name called out in the specification or drawing for a given pin (e.g., Random, Source, Reference, etc.).
ACCEPT LO:	The lower limit of a pin, V_n or I_n. Refer to definition table XXVI.
ACCEPT HI:	The upper limit of a pin, V_n or I_n. Refer to definition table XXVI.
Open/Ground:	If a given pin is a ground or an open (no connection, three-state) o = open, G = ground.
LOAD:	When a required load is connected to a pin it will be referenced in this column as Rn (refer to load connection table XXVII for points of connection).
Input:	Indicates a pin is an input (X).
Output:	Indicates a pin is an output (*). NOTE: All output data shall be recorded.
INIT:	Condition of a pin at the onset of a test (all input/output statements remain in defined states until changed within text. Also when INIT: reappears, the original conditions as defined in TIC table XXIX are re-established).

B.3.5 Definition of symbols. Table XXX is a definitive listing of all symbols which will be encountered in the body of the SSS.

TABLE XXX. Symbols.

Symbol	Name	Description
*	Output pin	Designated pins preceded by an "*" are output pins.
V_n	Voltage	V_n, dc V, unless otherwise specified.
F	Frequency	The measured rate of a periodic signal, sine wave, Hz, V_{pp}, unless otherwise specified. Example: *P1-G/F_3: Output signal at P1-G shall be 3.0 $\pm$ 1.8 V_{pp} at a frequency of 1.5 $\pm$ 0.8 Hz (see table XXVI).
I	Current	dc mA, unless otherwise defined.
HIn	High	HI level, voltage or current, as defined in the TIC (table XXIX). Note: A HIa notation refers to V_{na} as specified in the TIC, a HIb notation refers to V_{nb} in the same table. (This notation is used only when more than one voltage or current level is specified for a signal I.D.).
LOn	Low	LO level, voltage or current as defined in TIC (table XXIX). See note above.
o	Open	Open circuit, no connection (high impedance/three-state).
G	Ground	Continuity to ground, less than 1 ohm.
Rn	Resistance	Resistor value as defined in load table XXVII.

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APPENDIX B

TABLE XXX. Symbols - Continued.

Symbol	Name	Description
U	Undefined	State not specified.
#	Momentary	Stimulus (V, I) is momentarily applied.
±	Tolerance	Required if tolerance is not defined in tables XXVI, XXVII, XXVIII or XXIX.
/	Shall be	For output statements symbol means "shall be". For input statements symbol means "shall be connected to".
&	And	Separates simultaneous statements under the same sequence number.
:	End	End of operation(s) for a given sequence number. There shall be a minimum of 1 millisecond (ms) delay before performing the next sequence operation.
::	Hold	Delays further execution until the statement immediately following it is true, then continues.
P	Period	A pulse duration, in ms, unless otherwise specified. Example: J1-a/#HI(P100): Input pin J1-a shall be momentarily 'HI' for a period of 100 ms.
T_n	Time	An elapsed time duration between succeeding events, in ms, unless otherwise specified. Examples: a) $T_1 = 0$ & *J1-b/HI: b) $T_1 = 50$ & *J1-b/LO: T1 is not a special time so it is in ms. a) Output J1-b shall be a defined HI (see TIC) at time zero (T1 = 0). b) Output J1-b shall be a defined LO (see TIC) at a time 50 (ms) of T1.
J	Perform sequence(s)	Branching instruction, go to sequence number (defined), execute through sequence number (defined), and return to the next line in the sequence. Example: a) J/xx...nn b) J3-n/HI: (next statement) Branch shall go to sequence xx, execute through (...) sequence nn, then return to next statement in sequence after J statement. If only one sequence number is stated, perform that sequence and return to next statement.
ARRAYn	Array	Arrays are groups of input or output pins to be treated as a digital word, in the form of binary, hex, gray code, etc., and the defined HI/LO of TIC. Arrays are special conditions and will be defined in table XXVI.

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APPENDIX B

TABLE XXX. Symbols - Continued.

<table>
<tr><th>Symbol</th><th>Name</th><th>Description</th></tr>
<tr><td rowspan="2">%</td><td rowspan="2">Array sequence

Ramp</td><td>The array defined in definition table XXVI, sequenced through defined HI/LO's.
Example: Array A/10...25, J13:
Array A shall be input "word" 10, 11,... through input "word" 25, dwelling at least 10 ms at each "word", as shown below.
<table>
<tr><th rowspan="2">Array Input Word</th><th colspan="8">J13 Pin</th></tr>
<tr><th>K</th><th>J</th><th>H</th><th>G</th><th>F</th><th>E</th><th>D</th><th>C</th></tr>
<tr><td>10</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>11</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>.
.</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>25</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
</table>
NOTE: 1 = HI, 0 = LO</td></tr>
<tr><td>Linear voltage, current or resistance changing from one state to another over a fixed period of time.
Example: J1-a % HI(P100-P500):
J1-a shall ramp (linear change) to a defined HI from a previously defined state, within 100 and 500 ms.</td></tr>
</table>

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